# Lab Exercise 4

Lab Exercise: Selection Statements (`Switch..Case`) in C
Duration: 2 Hours

### Purpose of the Exercise

This exercise is designed to help students understand and apply the concept of selection control structures, specifically the switch-case statement in C programming. The tasks reinforce how decisions are made based on user input without using looping constructs.

### Learning Objectives

• Use the switch-case statement to control program flow based on variable values.
• Apply scanf and printf functions correctly for input and output operations.
• Develop simple application scenarios that require conditional branching.
• Understand how to handle invalid input using default case handling.

## Task 1 : Using switch..case Statement

### Exercise 3.1

Write a C program that reads an integer between 1 and 7 and prints the corresponding day of the week. Use a `switch..case` statement.

Example Output:

```
Enter day number (1-7): 3
Wednesday
```

### Exercise 3.2

Write a C program that takes a character input (A, B, C, D, F) and displays the grade description using `switch..case`.

Grade Table:
A – Excellent
B – Good
C – Average
D – Poor
F – Fail

Sample Output:

```
Enter grade: B
Good
```

## Task 2 : Basic Menu Selection

Write a C program that displays the following menu:

```
1. Add
2. Subtract
3. Multiply
4. Divide
```

Ask the user to enter an option (1–4). Using a `switch-case`, input two integers using scanf, then perform and display the correct operation.

Sample Output:

```
1. Add
2. Subtract
3. Multiply
4. Divide

Enter your choice (1-4): 1
Enter first numbers    : 3
Enter second numbers   : 7
Result                 : 10.00
```

```
Answer:
#include <stdio.h>
int main() {
    int choice;
    float num1, num2, result;

    printf("1. Add\n");
    printf("2. Subtract\n");
    printf("3. Multiply\n");
    printf("4. Divide\n");

    printf("Enter your choice (1-4): ");
    scanf("%d", &choice);

    printf("Enter first numbers: ");
    scanf("%f",&num1);

    printf("Enter second numbers: ");
    scanf("%f",&num2);


    switch(choice) {
        case 1:
            result = num1 + num2;
```

```
            printf("Result = %.2f\n", result);
            break;
        case 2:
            result = num1 - num2;
            printf("Result = %.2f\n", result);
            break;
        case 3:
            result = num1 * num2;
            printf("Result = %.2f\n", result);
            break;
        case 4:
            if(num2 == 0)
                printf("Error: Division by zero not allowed.\n");
            else {
                result = num1 / num2;
                printf("Result = %.2f\n", result);
            }
            break;
        default:
            printf("Invalid choice.\n");
    }
    return 0;
}
```

### Task 3: Character Grade Interpretation

Write a C program that reads a single character from the user:

```
Enter a grade (A, B, C, D):
```

Use a **switch-case** to print the meaning:

- A → "Excellent"
- B → "Good"
- C → "Satisfactory"
- D → "Pass"

If the user enters any other character, display `"Invalid grade"`.

Sample Output :

```
Enter a grade (A, B, C, D): B    (underline value enter by user)
Good
```

```
Answer:
#include <stdio.h>
int main() {
```

```
    char grade;

    printf("Enter a grade (A, B, C, D): ");
    scanf(" %c", &grade);

    switch(grade) {
        case 'A': printf("Excellent\n"); break;
        case 'B': printf("Good\n"); break;
        case 'C': printf("Satisfactory\n"); break;
        case 'D': printf("Pass\n"); break;
        default: printf("Invalid grade\n");
    }
    return 0;
}
```

## Task 4: Restaurant Ordering System

A restaurant offers a menu. Ask user for choice and display the price using switch-case.

A restaurant offers the following items:

```
1. Nasi Goreng       RM6.50
2. Mee Goreng        RM6.00
3. Teh Ais           RM2.00
4. Milo Ais          RM2.50
```

Write a C program that:

1. Displays the menu.
2. Prompts the user to enter their choice using `scanf`.
3. Uses a **switch-case** to print the price of the selected item.
4. If the choice is not 1–4, print `"Invalid menu selection"`.

Sample expected output:

```
Enter menu number: 1
You selected menu  : Nasi Goreng
Price              : RM6.50
```

```
Answer:
#include <stdio.h>
int main() {
    int choice;

    printf("1. Nasi Goreng       RM6.50\n");
    printf("2. Mee Goreng        RM6.00\n");
    printf("3. Teh Ais           RM2.00\n");
    printf("4. Milo Ais          RM2.50\n");
```

```
    printf("Enter menu number: ");
    scanf("%d", &choice);

    switch(choice) {
        case 1: printf("You selected Nasi Goreng. Price =
RM6.50\n"); break;
        case 2: printf("You selected Mee Goreng. Price =
RM6.00\n"); break;
        case 3: printf("You selected Teh Ais. Price = RM2.00\n");
break;
        case 4: printf("You selected Milo Ais. Price =
RM2.50\n"); break;
        default: printf("Invalid menu selection\n");
    }
    return 0;
}
```